

MUHAMMAD ZOHAIB

Full Stack Developer | TypeScript · Node.js · React.js · Next.js · React Native ·

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Portfolio | <https://craft-code-showcase-11.vercel.app/>

PROFESSIONAL SUMMARY

Motivated, detail-oriented Full Stack Developer with professional experience architecting and delivering scalable web and mobile applications for production environments. Experienced in designing and managing relational databases (MySQL) using Sequelize ORM — including schema design, query optimization, and data relationships for complex, high-volume systems. Skilled in building Node.js microservices architectures, with services communicating through queue-based systems (BullMQ/Redis) instead of direct HTTP calls, following clean service layering (API gateway → business logic → data layer) for scalable, decoupled backend systems.

Equally proficient in building high-performance, SEO-optimized web interfaces using React.js and Next.js with SSR and SSG, and engineering cross-platform mobile experiences with React Native — with TypeScript used consistently across every professional project. Experienced in AI-powered chatbot and voice bot development, including building conversational engines using Retrieval-Augmented Generation (RAG) for context-aware responses and integrating ElevenLabs for natural, real-time voice interaction. Strong focus on application performance through multi-layer caching strategies (sessionStorage, Redux, and query-level optimization), component memoization using React.memo, and scalable global state management with Redux Toolkit (createSlice, createAsyncThunk, RTK Query).

Demonstrates specialized expertise in enterprise-grade authorization and access control, leveraging Casbin for RBAC, ABAC, and ACL enforcement, integrated across both backend middleware and frontend route protection. Committed to clean architecture, scalable system design, agile collaboration, and delivering production-ready, full-stack software that exceeds business requirements.

CORE COMPETENCIES

- **Full-Stack Development & Backend Architecture** (TypeScript, Node.js, React.js, Next.js)
 - **Database Management & API Design**
 - **Frontend & Mobile Development**
 - **Performance Optimization & Collaboration**
 - with microservices architecture using queue-based systems (BullMQ/Redis) and clean service layering
 - RESTful API design, MySQL database management with Sequelize ORM (schema design, query optimization), and enterprise-grade authorization using Casbin (RBAC/ABAC/ACL)
 - RESTful API Design & MySQL Database ManagementCross-platform mobile apps with React Native, SSR/SSG web interfaces with Next.js for SEO & performance, and frontend middleware for route permission handling
 - Multi-layer caching, memoization (React.memo), scalable state management (Redux Toolkit), version control, and agile collaboration
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PROFESSIONAL EXPERIENCE

| Full Stack Developer

2025 – Present | Lahore, Pakistan

- As a Full Stack Developer, I've architected and shipped production-grade systems that combine robust backend engineering with polished, high-performance user experiences. I engineered an enterprise-grade authorization layer using Casbin, enforcing role- and attribute-based access control (RBAC/ABAC) across every API endpoint and data layer — then extended that security model into the frontend by building custom permission middleware that dynamically renders UI and intercepts navigation in real time, closing the gap between backend rules and what users actually see.
- I spearheaded the end-to-end development of an AI-powered mobile assistant for iOS and Android using React Native — from architecture and feature design through to launch. I built its conversational core using Retrieval-Augmented Generation (RAG) for context-aware responses, and integrated ElevenLabs to power natural, real-time voice interaction, turning a standard app into an intelligent, voice-enabled product. I also optimized its performance using React.memo, useMemo, and useCallback, cutting redundant re-renders and delivering a noticeably smoother mobile experience.
- On the data layer, I designed normalized MySQL schemas and optimized complex queries to keep high-traffic features fast and reliable, while building and maintaining RESTful APIs in Node.js with clean architecture, input validation, and structured error handling. I structured scalable global state using Redux Toolkit — leveraging createSlice, createAsyncThunk, and RTK Query — to keep both web and mobile applications in sync with the server efficiently.
- I enforce strict TypeScript across every project I build, using custom interfaces and generics to eliminate runtime errors and keep large codebases maintainable as they scale — writing code that's not just functional, but built to last.

TECHNICAL SKILLS

Backend	Node.js (TypeScript), Express.js, Next.js API Routes, RESTful API Design & Development, Microservices Architecture, BullMQ/Redis (Queue-Based Communication), Sequelize ORM, JWT Authentication
Database	MySQL — Schema Design, Query Optimization, Data Modeling, Sequelize ORM
AI/Automation	AI Integration, Retrieval-Augmented Generation (RAG), ElevenLabs (Voice AI), Chatbot & Voice Bot Development, AI Content Generation (Text-to-Image, Caption Editing)
Cloud/Real-time Services	Firebase Authentication, Firebase Cloud Messaging (FCM)
Frontend	TypeScript, React.js, Next.js (SSR/SSG), Redux Toolkit (RTK Query, createSlice, createAsyncThunk), JavaScript (ES6+), HTML5, CSS3
Tools & DevOps	Git, GitHub, Postman, VS Code, Cursor AI, Claude
Methodology	Agile / Scrum, Code Review, Clean Architecture, Modular Design Patterns

KEY PROJECTS & ACHIEVEMENTS

MoboCheck Web Platform — Professional Project | Mobocheck

Tech Stack: TypeScript, React.js, Next.js, Node.js, Casbin, Redux Toolkit, MySQL

- Contributed to the design and development of a live, user-facing production web platform, delivering features end-to-end from UI through database layer
- Implemented Casbin-powered authorization to enforce role-specific access across all platform modules and REST API endpoints, significantly strengthening application security
- Leveraged Next.js SSR to improve initial page load performance and ensure search engine indexability across key application routes

QYRA — AI Assistance Mobile Application — Professional Project | Mobocheck

Tech Stack: TypeScript, React Native, Node.js, Redux Toolkit, REST APIs

- Led full mobile development lifecycle for a cross-platform AI-powered application targeting both iOS and Android, from architecture through production-ready delivery
- Implemented React.memo and component-level memoization to minimize unnecessary renders, resulting in noticeably improved runtime performance and responsiveness on mobile devices
- Integrated AI-powered backend services via RESTful APIs, managing asynchronous data flows, error boundary handling, and graceful degradation for a seamless user experience

Other Projects:

- **Postsiva** — AI-Powered Social Media Management Platform (Automatev)
- **Event Management System** — Academic Project (University of Gujrat)
- **Freelance Marketplace Hub** — Personal/Learning Project

EDUCATION

University of Gujrat

Bachelor of Science — *Information Technology*

2021 – 2025 | Gujrat

Software Engineering • Full-Stack Development • Database Systems

PROFESSIONAL ATTRIBUTES

- Strong written and verbal communication skills, with proven ability to collaborate effectively across remote, cross-functional development teams
- Skilled at translating ambiguous or evolving requirements into well-structured, scalable software solutions
- Systematic problem-solver with hands-on experience in debugging, root cause analysis, and resolving production issues under time pressure
- Proactive learner who stays current with emerging technologies — recently applied this by integrating AI and voice technologies (RAG, ElevenLabs) into production mobile applications
- Comfortable working in Agile/Scrum environments, contributing to sprint planning, daily standups, and collaborative code reviews